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Uncertainty Principle of Ambiguity Function in Linear Canonical Transform Domain

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***Abstract*—The ambiguity function (AF) and its related uncertainty function are essential time-frequency analysis method for the waveform properties and echo-location in radar and optical applications. This paper devotes to investigate new uncertainty principles of AF in linear canonical transform domain (AFL) for complex-value function. Firstly, the moments of AFL are defined to measure mean and the covariance of signal in time and frequency domain. Then new uncertainty principles of AFL are proposed related to moments, which capture lower bounds for product of time delay and Doppler spreads. The lower bound can be only achieved by Gaussian-type function. Furthermore, an example with Gaussian-type function is performed to verify the correctness of proposed uncertainty principles. Finally, potential applications of derived theorems are explored in the field of radar.**

Keywords - Uncertainty principle; Linear canonical transform; Ambiguity function; Complex-value function; Moments

I. INTRODUCTION

The classical Heisenberg's uncertainty principle based on Fourier transform (FT) reveals a lower bound for product of time and frequency spread of a function, which can not both be localized sharply. The uncertainty principle is of great importance and provides a vital theoretical foundation in many fields such as the quantum mechanics, optical fiber, radar application and other fields of information processing [1-3].

Indeed, from its first statement in time and FT domain to its newest developments, the uncertainty principle has been encountered many parallel evolutions and generalizations in different domains. Linear canonical transform (LCT) is the generalized integral transform with four parameters and more freedom, which can be reduced to FT, fractional FT (FRFT) and other transforms. LCT has been widely well-known and used in time frequency analysis for signal with time-varying statistical characteristics [4]. Recently, the uncertainty principles in LCT domain have born much fruits related to various notions or time LCT-frequency representations [5-8]. Since LCT can not support the local frequency information of signal due to its global kernel function, some efforts have already been put into windowed LCT, Wigner Ville distribution, ambiguity function (AF) and other time frequency representation methods in LCT domain [9,10]. Correspondingly, some uncertainty inequalities have been proposed [11-13].

The AF with better time delay and Doppler frequency resolution was introduced into LCT domain (AFL), which receives much attention to analyze non-stationary signal and waveform characteristics in the radar and other applications. However, there are few studies about uncertainty principles related to AFL. The literature [11] gave a lower bound of product of time-delay and Doppler spread, which was depicted by logarithmic uncertainty relation of AFL for a odd or even signal. Another literature [12] investigated the uncertainty principle of a generalized time-frequency transform in FRFT domain by the light of classical uncertainty principle, but there is no the condition to be discussed that equality holds. It needs to be noticed that above results have not involved higher-order moments for AFL. As a matter of fact, moments can describe a variety of the statistical characteristics of signal including time and Doppler frequency spread. Therefore, it is meaningful and worthwhile to study uncertainty relationship of time and Doppler spread in AFL domain based on moments for signal analysis and processing.

This paper is to establish the uncertainty principles of AFL for complex-value function and derive the lower bound based on higher-order moments. The paper is organized as follows: Section II is Preliminary. Section III presents the new definition of different moments for AFL. The uncertainty principles associated with the AFL are investigated. Section IV discusses the potential application for proposed result. Conclusion is drawn in Section V.

II. PRELIMINARY

A. Linear canonical transform

Definition 1. For signal $x(t)=|x(t)|e^{j\phi(t)}\in L^2(\mathbb{R})$, $tx(t)\in L^2(\mathbb{R})$ let $A=(a,b,c,d)$ be a real parameter matrix and $|A|=ad-bc=1$. The LCT of $x(t)$ is defined as

$$L_A^x(u)=L_A[x(t)](u)=\int_{\mathbb{R}}x(t)K_A(t,u)\mathrm{d}t \tag{1}$$

with kernel function

$$K_A(t,u)=\begin{cases}\dfrac{1}{\sqrt{j2\pi b}}e^{j(\frac{a}{2b}t^2-\frac{1}{b}tu+\frac{d}{2b}u^2)}, b\neq 0\\ \sqrt{d}e^{j\frac{cd}{2}u^2}\delta(t-du), b=0\end{cases} \tag{2}$$

where L_A denotes LCT operator [4].

The inverse LCT operator L_A^{-1} is

$$x(t)=L_A^{-1}\left\{L_A^x(u)\right\}=\int_{\mathbb{R}}L_A^x(u)K_{A^{-1}}(u,t)\mathrm{d}u \tag{3}$$

where $K_{A^{-1}}(u,t)$ is the kernel function of inverse LCT and $A^{-1}=(d,-b,-c,a)$. In fact, kernel function shares unitarity $K_A^*(t,u)=K_{A^{-1}}(u,t)$.

Remark 1: Since the LCT with b=0 reduces to be a kind of scaling and chirp multiplication operations, this paper will only discuss $b\neq 0$.

B. Ambiguity function in LCT domain

In order to acquire better time-frequency resolution and time-frequency location, AF is introduced into LCT domain to analyze complex-value and non-stationary signal and system.

Definition 2. For signal $x(t)=|x(t)|e^{j\phi(t)}\in L^2(\mathbb{R})$, the AFL of signal $x(t)$ is defined as

$$AFL(\tau,v)=\frac{1}{\sqrt{2\pi}}\int_{\mathbb{R}}x(t+\frac{\tau}{2})x^*(t-\frac{\tau}{2})K_A(t,v)\mathrm{d}t \tag{4}$$

where τ and v denote time delay regarding to t and Doppler frequency in LCT domain, $K_A(t,v)$ is transform kernel function in LCT domain.

Lamma 1: For signal $x(t)=|x(t)|e^{j\phi(t)}\in L^2(\mathbb{R})$, there are

$$\|AFL(\tau,v)\|^2=2\|x(m)\|^2\|x^*(n)\|^2. \tag{5}$$

where $m=t+\frac{\tau}{2}, n=t-\frac{\tau}{2}$ and $\|x(t)\|^2=\int_{\mathbb{R}}|x(t)|^2\mathrm{d}t$.

Proof: According to the definition of AFL in Eq (4),

$$\begin{aligned}\|AFL(\tau,v)\|^2&=\int_{\mathbb{R}}\int_{\mathbb{R}}|AFL(\tau,v)|^2\mathrm{d}\tau\mathrm{d}u\\&=\frac{1}{2\pi}\iint_{\mathbb{R}}\left[\int_{\mathbb{R}}x(t_1+\frac{\tau}{2})x^*(t_1+\frac{\tau}{2},t_1-\frac{\tau}{2})K_A(t_1,v)\mathrm{d}t_1\right]\\&\quad\left[\int_{\mathbb{R}}x(t_2+\frac{\tau}{2})x^*(t_2-\frac{\tau}{2})K_A(t_2,v)\mathrm{d}t_2\right]^*\mathrm{d}\tau\mathrm{d}u\\&=\int_{\mathbb{R}}\int_{\mathbb{R}}\left[x(t_1+\frac{\tau}{2})x^*(t_1+\frac{\tau}{2})\right]\left[x(t_1-\frac{\tau}{2})x^*(t_1-\frac{\tau}{2})\right]\mathrm{d}t_1\mathrm{d}\tau\end{aligned} \tag{6}$$

Let $m=t_1+\frac{\tau}{2}, n=t_1-\frac{\tau}{2}$, $\mathrm{d}t_1=\mathrm{d}n, \mathrm{d}\tau=2\mathrm{d}m$, then

$$\begin{aligned}\|AFL(\tau,v)\|^2&=2\int_{\mathbb{R}}\int_{\mathbb{R}}\left[x(m)x^*(m)\right]\left[x(n)x^*(n)\right]^*\mathrm{d}m\mathrm{d}n\\&=2\int_{\mathbb{R}}\int_{\mathbb{R}}|x(m)|^2|x^*(n)|^2\mathrm{d}m\mathrm{d}n\\&=2\|x(m)\|^2\|x^*(n)\|^2.\end{aligned} \tag{7}$$

The proof is finished.

C. Moment and uncertainty principle

This subsection will present some moments of signal, in addition to uncertainty principle of LCT related to moments. On account of Parseval formula, signal energy holds in different domain.

Definition 3. Let signal $x(t)=|x(t)|e^{j\phi(t)}\in L^2(\mathbb{R})$, then there are first moment and second moment as follows [7,8]

$$\begin{aligned}\overline{t_x}&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}t|x(t)|^2\mathrm{d}t\\ \overline{w_x}&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}w|X(w)|^2\mathrm{d}w\\ \overline{u_x}&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}u|L_A^x(u)|^2\mathrm{d}u\end{aligned} \tag{8}$$

$$\begin{aligned}T_x^2&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}(t-\overline{t_x})^2|x(t)|^2\mathrm{d}t\\ F_x^2&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}(w-\overline{w_x})^2|X(w)|^2\mathrm{d}w\\ F_{A,x}^2&=\frac{1}{\|x(t)\|^2}\int_{\mathbb{R}}(u-\overline{u_x})^2|L_A^x(u)|^2\mathrm{d}u\end{aligned} \tag{9}$$

where $X(w)$ is the FT of signal $x(t)$, $\overline{t_x}$, $\overline{w_x}$, $\overline{u_x}$ are the first moments (mean) of $x(t)$ in the time, frequency and LCT domains respectively. $T_x^2, F_x^2, F_{A,x}^2$ are the second moment (variance) of $x(t)$ corresponding to the spreads in the time, frequency and LCT domains, respectively. These moments are qualified to describe the statistical and distribution information of signal.

Lamma 2. The uncertainty principles of the LCT based on moments as [5]

$$T_x^2F_x^2\geq b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+\left(aT_x^2+bCov_x\right)^2 \tag{10}$$

where covariance Cov_x and absolute covariance COV_x of $x(t)$ have the forms with phase derivation $\phi'(t)$

$$Cov_x = \int_{\mathbb{R}} (t-\overline{t_x})^2 (\phi'(t) - \overline{w_x}) \,|\, x(t) \,|^2 \mathrm{d}t$$
$$COV_x = \int_{\mathbb{R}} \left| (t-\overline{t_x})^2 (\phi'(t) - \overline{w_x}) \right| |\, x(t) \,|^2 \mathrm{d}t. \tag{11}$$

The uncertainty principles of LCT in Lemma 2 gives shaper lower bounds, which is able to establish a theoretical foundation for complex signal analysis as well as practical application.

III. Uncertainty Principle for AFL

This Section will propose the moment and uncertainty principles of AFL, which helps to analyze easily local useful feature of complex signal.

A. Moment of AFL

Definition 4. Let signal $x(t) = |x(t)| e^{j\phi(t)} \in L^2(\mathbb{R})$, then the first and second moments of AFL are defined as

$$\overline{\tau_{A,AF}} = \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \tau \,|\, AFL_x^A(\tau, v) \,|^2 \mathrm{d}\tau \mathrm{d}v \tag{12}$$

$$\overline{v_{A,AF}} = \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} v \,|\, AFL_x^A(\tau, v) \,|^2 \mathrm{d}\tau \mathrm{d}v \tag{13}$$

$$T_{A,AF}^2 = \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \left(\tau - \overline{\tau_{A,AF}} \right)^2 |\, AFL_x^A(\tau, v) \,|^2 \mathrm{d}\tau \mathrm{d}v \tag{14}$$

$$F_{A,AF}^2 = \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \left(v - \overline{v_{A,AF}} \right)^2 |\, AFL_x^A(\tau, v) \,|^2 \mathrm{d}\tau \mathrm{d}v \tag{15}$$

where $\overline{\tau_{A,AF}}$, $\overline{v_{A,AF}}$ are the first moments (mean) of $x(t)$ in the time and AFL domains, respectively. $T_{A,AF}^2$, $F_{A,AF}^2$ are the second moments (spread) in the time and AFL domains, respectively.

Lemma 3. Let signal $x(t) = |x(t)| e^{j\phi(t)} \in L^2(\mathbb{R})$, there are

$$\overline{\tau_{A,AF}} = 0 \tag{16}$$
$$\overline{v_{A,AF}} = 0 \tag{17}$$

Proof: Based on the Eq. (5) and (12), we have

$$\begin{aligned}\overline{\tau_{A,AF}} &= \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \tau \,|\, AFL_x^A(\tau, v) \,|^2 \mathrm{d}\tau \mathrm{d}v \\ &= \frac{1}{\left\| AFL_x^A(\tau, v) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} \left[x(t_1 + \frac{\tau}{2}) x^*(t_1 + \frac{\tau}{2}) \right] \\ &\qquad \left[x(t_1 - \frac{\tau}{2}) x^*(t_1 - \frac{\tau}{2}) \right] \mathrm{d}t_1 \mathrm{d}\tau\end{aligned} \tag{18}$$

Let $m = t_1 + \frac{\tau}{2}, n = t_1 - \frac{\tau}{2}, \mathrm{d}t_1 = \mathrm{d}n, \mathrm{d}\tau = 2\mathrm{d}m.$ Then

$$\begin{aligned}\overline{\tau_{A,AF}} &= \frac{1}{\left\| x(m) \right\|^2 \left\| x^*(n) \right\|^2} \int_{\mathbb{R}} \int_{\mathbb{R}} (m-n) \,|\, x(m) \,|^2 |\, x^*(n) \,|^2 \mathrm{d}m \mathrm{d}n \\ &= \frac{1}{\left\| x(m) \right\|^2} \int_{\mathbb{R}} m \,|\, x(m) \,|^2 \mathrm{d}m - \frac{1}{\left\| x^*(n) \right\|^2} \int_{\mathbb{R}} n \,|\, x^*(n) \,|^2 \mathrm{d}m \\ &= \overline{t_x} - \overline{t_x} = 0.\end{aligned} \tag{19}$$

Similarly, Eq. (13) can be derived by same method. The proof is finished.

Lemma 3. states that means of time delay τ and Doppler frequency shift v locate at the center in the time delay and LCT frequency plane, which helps derive uncertainty principle of AFL.

B. Uncertainty principle of AFL

Theorem 1. For signal $x(t) = |x(t)| e^{j\phi(t)} \in L^2(\mathbb{R})$, $tx(t) \in L^2(\mathbb{R})$ let $A = (a, b, c, d)$ be a real parameter matrix and $|A| = ad - bc = 1$, then

$$T_{A,AF}^2 F_{A,AF}^2 \ge 4b^2 \left(\frac{1}{4} + COV_x^2 - Cov_x^2 \right) + 4 \left(a T_x^{\,2} + b Cov_x \right)^2 \tag{20}$$

The equality holds if and only if $x(t)$ has one of the following four Gaussian-type functions

$$\begin{aligned}
x(t) &= \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_1 \right]}, \\
x(t) &= \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[-\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_2 \right]}, \\
x(t) &= \begin{cases} \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_3 \right]}, t \ge \overline{t_x} \\ \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[-\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_4 \right]}, t < \overline{t_x} \end{cases} \\
x(t) &= \begin{cases} \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[-\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_5 \right]}, t \ge \overline{t_x} \\ \left(\frac{\gamma}{\pi} \right)^{\frac{1}{4}} e^{-\frac{\gamma}{2}(t-\overline{t_x})^2} e^{j\left[\frac{\gamma}{2}(t-\overline{t_x})^2 + \overline{w_x} t + \rho_6 \right]}, t < \overline{t_x} \end{cases}
\end{aligned} \tag{21}$$

where $\gamma, \lambda > 0$, and $\rho_i \in \mathbb{R}, i = 1, 2, \cdots, 6$.

Proof: First, given the definition in Eq. (14), we have

$$T_{A,AF}^2 = \frac{1}{\left\|AFL_x^A(\tau,v)\right\|^2}\int_{\mathbb{R}}\int_{\mathbb{R}}\left(\tau-\overline{\tau_{A,AF}}\right)^2 |AFL_x^A(\tau,v)|^2 \mathrm{d}\tau\mathrm{d}v$$

$$= \frac{1}{\left\|AFL_x^A(\tau,v)\right\|^2}\int_{\mathbb{R}}\int_{\mathbb{R}}\left(\tau-\overline{\tau_{A,AF}}\right)^2 \left|x(t_1+\frac{\tau}{2})\right|^2 \left|x^*(t_1-\frac{\tau}{2})\right|^2 \mathrm{d}t_1\mathrm{d}\tau \tag{22}$$

Let $m = t_1+\frac{\tau}{2}, n = t_1-\frac{\tau}{2}, \tau = m-n$,Then

$$T_{A,AF}^2 = \frac{1}{\left\|x(m)\right\|^2\left\|x^*(n)\right\|^2}\int_{\mathbb{R}}\int_{\mathbb{R}}(m-n)^2 |x(m)|^2 |x^*(n)|^2 \mathrm{d}m\mathrm{d}n$$

$$= \frac{1}{\left\|x(m)\right\|^2}\int_{\mathbb{R}} m^2 |x(m)|^2 \mathrm{d}m - \frac{1}{\left\|x^*(n)\right\|^2}\int_{\mathbb{R}} n^2 |x^*(n)|^2 \mathrm{d}m$$

$$-\frac{2\int_{\mathbb{R}} m|x(m)|^2\mathrm{d}m\int_{\mathbb{R}} n|x^*(n)|^2\mathrm{d}m}{\left\|x(m)\right\|^2\left\|x(n)\right\|^2}$$

$$= T_x^2 + T_{x^*}^2 - 2\overline{\tau_{A,AF}}$$

$$= 2T_x^2 \tag{23}$$

Similarly,

$$F_{A,AF}^2 = 2F_{A,x}^2 \tag{24}$$

Thus,

$$T_{A,AF}^2 F_{A,AF}^2 = 4T_x^2 F_{A,x}^2$$

$$\geq 4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x{}^2+bCov_x\right)^2 \tag{25}$$

The proof is finished.

Theorem 1 gives the product of time delay and Doppler spreads by uncertainty inequality. It not only shows there is the lower bound of time- frequency spread in AFL domain, but also means that time delay resolution and Doppler resolution can not both sharply localized for general radar signal but special Gaussian-type signal. Parameter *a* and *b* in parameter matrix of LCT will affect the lower bound of time-frequency in AFL domain. As for frequency modulation signal, its effective frequency spread in AFL domain could be smaller when the parameter chosen appropriate or optimal parameters. Theorem 1 can be reduced in FRFT domain and FT domain when parameter matrix is special.

Corollary 1. For signal $x(t) = |x(t)|e^{j\phi(t)} \in L^2(\mathbb{R})$, $tx(t) \in L^2(\mathbb{R})$, the uncertainty principle of AF in FRFT domain is derived with $A = (\cos\alpha, \sin\alpha, -\sin\alpha, \cos\alpha),$, $\alpha \in \mathbb{R}$, then

$$T_{\alpha,AF}^2 F_{\alpha,AF}^2 \geq$$
$$4\sin\alpha^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(\cos\alpha T_x{}^2+\sin\alpha Cov_x\right)^2 \tag{26}$$

Corollary 2. For signal $x(t) = |x(t)|e^{j\phi(t)} \in L^2(\mathbb{R})$, $tx(t) \in L^2(\mathbb{R})$,the uncertainty principle of classical AF is derived with $A = (0,1,-1,0)$, then

$$T_{\alpha,AF}^2 F_{\alpha,AF}^2 \geq 1+4COV_x^2. \tag{27}$$

Remark 2: In view of the relationship AF and Wigner-Valle distribution in LCT domain, it is worth noting that the similar discussion and uncertainty principle can be derived for WVD in LCT domain.

C. Comparison and explanation

The short-time LCT (STLCT) is a novel time-frequency analysis tool, which is known as windowed linear canonical transform and strengthen the traditional the LCT. Its uncertainty principle has been studied to state that the product of the variances of the signal in the time and frequency domains has a lower bound [6].

For complex signal $x(t) = |x(t)|e^{j\phi(t)} \in L^2(\mathbb{R})$, $tx(t) \in L^2(\mathbb{R})$, the uncertainty principles derived by STLCT [6] and AFL in this paper are shown in Table 1. Compared to the result of STLCT, although proposed uncertainty principle of AFL shares four times that of the first two terms of STLCT, the result of STLCT includes $b^2\left(\frac{1}{4}+COV_x^2\right)$. Thus, parameter *b* affects sharply lower bound of product of time and LCT frequency spread for STLCT and AFL.

Table1 Comparison with STLCT

Method	Result
STLCT [6]	$T_{A,S}^2F_{A,S}^2 \geq$ $b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+\left(aT_x{}^2+bCov_x\right)^2+b^2\left(\frac{1}{4}+COV_x^2\right)$ (28)
AFL [this paper]	$T_{A,AF}^2F_{A,AF}^2 \geq$ $4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x{}^2+bCov_x\right)^2$

Moreover, regarding the complexity for the product of time-frequency (Doppler) spread, one finds the fact that proposed uncertainty relationship Eq (20) in AFL domain is same to is same to Eq. (10) in LCT domain [5] and Eq. (28) in STLCT domain [6].

IV. Potential Application

The AFL illustrates the distribution of time delay and Doppler frequency in LCT domain, the superior detection and estimation performance has been verified in the noise environment. This section will discuss the potential application of uncertainty principle of AFL in signal processing, which further shows the importance of proposed theorems. The philosophy of uncertainty principles of AFL are the same as that for the uncertainty principles in the LCT setting.

The uncertainty principle of AFL indicates the relationship between time-delay and Doppler spreads by a lower bound, which are generally used to evaluate effective Doppler spectra in affine modulation systems. For proposed uncertainty principles of AFL, if $T_{A,AF}$ is known, then

$$\begin{aligned} F_{A,AF}^2 &\geq \frac{4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x^{\ 2}+bCov_x\right)^2}{T_{A,AF}^2} \\ &\geq \frac{b^2}{T_{A,AF}^2} \end{aligned} \tag{29}$$

The first inequality of Eq. (29) can be reached until working signal $x(t)$ is one of four chirp signals with Gaussian-type amplitude given in Eq. (21). So it can be rewritten as

$$\begin{aligned} &4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x^{\ 2}+bCov_x\right)^2 \\ &=\min\left\{T_{A,AF}^2F_{A,AF}^2 : \int_{\mathbb{R}}|x(t)|^2\,\mathrm{d}t=1, tx(t)\in L^2(\mathbb{R})\right\} \end{aligned} \tag{30}$$

Eq. (30) shows that $\frac{4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x^{\ 2}+bCov_x\right)^2}{T_{A,AF}^2}$ can be reached by minimum value $T_{A,AF}^2F_{A,AF}^2$.

For a practical signal this minimum value is hard to reach, and the product of corresponding spreads is actually larger than $\frac{4b^2\left(\frac{1}{4}+COV_x^2-Cov_x^2\right)+4\left(aT_x^{\ 2}+bCov_x\right)^2}{T_{A,AF}^2}$.

For example, if a class of signals with $T_x \geq \sigma > 0$ is discussed in the practical environment with $A=(1,0,1,1)$, , then the uncertainty product cannot be less than $4\sigma^4$.

V. Conclusions

This paper formulates the uncertainty principles of AFL related to the covariance of time and frequency for complex signal. It makes sense to reveal the uncertainty relationship of product of time-delay and Doppler frequency spreads over AFL domain and states that both the time delay and Doppler can not be located sharply. The lower bound is obtained by proposed uncertainty principle and can be reached by a kind of chirp signal with Gaussian amplitude. Finally, the potential applications of proposed uncertainty principle is discussed and more potential limitations and practical applicability will be further studied in our future works.

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